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INVITED REVIEW

The athlete's heart: allometric considerations on published papers and relation to cardiovascular variables

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Abstract

To evaluate the morphology of the "athlete's heart", left ventricular (LV) wall thickness (WT) and end-diastolic internal diameter (LVIDd) at rest were addressed in publications on skiers, rowers, swimmers, cyclists, runners, weightlifters ($n = 927$), and untrained controls ($n = 173$) and related to the acute and maximal cardiovascular response to their respective disciplines. Dimensions of the heart at rest and functional variables established during the various sport disciplines were scaled to body weight for comparison among athletes independent of body mass. The two measures of LV were related ($r = 0.8$; $P = 0.04$) across athletic disciplines. With allometric scaling to body weight, LVIDd was similar between weightlifters and controls but 7%-15% larger in the other athletic groups, while WT was 9%-24% enlarged in all athletes. The LVIDd was related to stroke volume, oxygen pulse, maximal oxygen uptake, cardiac output, and blood volume ($r = \sim 0.9$, $P < 0.05$), while there was no relationship between WT and these variables ($P > 0.05$). In conclusion, while cardiac enlargement is, in part, essential for the generation of the cardiac output and thus stroke volume needed for competitive endurance exercise, an enlarged WT seems important for the development of the wall tension required for establishing normal arterial pressure in the enlarged LVIDd.

Keywords Athlete's heart · Blood volume · Endurance exercise · Left ventricular diameter · Left ventricular wall thickness · Cardiac adaptation · Resistance exercise · Scaling · Stroke volume

Introduction

The cardiovascular response to different sport disciplines (Caselli et al. 2016; Pressler et al. 2018) provides, besides information on the acute hemodynamic challenge to the heart, also a chronic stimulus for peripheral adaptations and cardiac structural remodeling. The influence of exercise training on the heart's adaptive structural remodeling was demonstrated initially in the rat (Secher 1921, 1923) and later confirmed in elite athletes, mainly by echocardiography at rest (Maron 1986; Spirito et al. 1994; Baggish et al. 2008; Utomi et al. 2013). Accordingly, the remodeling of the myocardium in response to long-lasting athletic participation has been characterized in two distinct categories by Morganroth et al. (1975). One category predominantly involves enlarged left ventricular internal diastolic diameter (LVIDd) for generating the high cardiac output (CO) required in endurance sport disciplines, and the other category involves enlarged LV wall thickness (WT) with no change in LVIDd, probably reflecting the high arterial pressure developed in power sport disciplines (MacDougall et al. 1985).

This dichotomous classification of the heart's structural properties has been adopted in the classification of sport disciplines based on their relative components of "dynamic" vs. "static" exercise (Mitchell et al. 2005; Levine et al. 2015). However, this dichotomous classification is at odds with observations that demonstrate concurrent enlarged LVIDd and WT in many athletes (Pluim et al. 2000; Utomi et al. 2013), although that is not a consistent finding (Baggish et al. 2008; Spence et al. 2011). These seemingly discrepant findings might be explained with consideration of Laplace's

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law suggesting that an enlarged LVIDd also requires an enlarged WT for pressure generation. In support, 7 months training increased LVIDd by 3.5% and WT by 11% in rowers, along with increased maximal oxygen uptake (VO_{2max}; Wieling et al. 1981). Conversely, short term (~4 week) detraining attenuated VO_{2max} as a result of a reduced blood volume (BV), stroke volume (SV), and CO (Mujika and Padilla 2000), while detraining for 6–34 weeks (Maron 1986) to ~5 years (Pelliccia et al. 2002) reduces VO_{2max} and remodels the heart to pre-training dimensions. Further evidence for a mixed concentric/eccentric left ventricular enlargement is provided by empirical studies in cross-trained subjects who both ran and weight trained (Elias et al. 1991) and alpine skiers (George et al. 1999). Taken together, the dichotomous classification of the morphologic modification of the heart in response to training needs revision to include "mixed" cardiac phenotypes that reflect the "mixed" cardiovascular load developed in the spectrum between the distinct "static" and "dynamic" categories of sport classification (Mitchell et al. 2005; Levine et al. 2015).

Comparison of the heart dimensions among participants in different sports is complicated by the effect of body size as recognized by Utomi et al. (2013). Differences in absolute cardiac dimensions, e.g., between men and women, can be normalized with scaling, as has been shown in Alpine skiers (George et al. 1999). Furthermore, it has been shown that allometric scaling is more efficient in normalizing cardiac dimensions than ratio-metric scaling (Dewey et al. 2008). In the present narrative review, data from published papers are used to relate LVIDd and WT to expansion of BV at rest (Heinicke et al. 2001; Zelenkova et al. 2019), and the maximum blood pressure to SV measured during sport-specific exercise in athletes. Also, data for the right heart and both left and right atria were included when available. The LV dimensions and cardiovascular variables were allometrically scaled to distinguish between the influence of training and body mass on the heart.

Methods

We included studies on athletes participating in endurance sports (skiing, rowing, swimming, cycling, and running) and weightlifting without discriminating between weight groups in, e.g., rowing and weightlifting and included sub-disciplines, i.e., middle- and long-distance running and cycling, different styles of swimming, skiing, 'Olympic' weightlifting, and resistance exercise (e.g., leg-press, deadlift, and squat). Athletes who engage in several exercise modalities were not included, e.g., triathletes and those participating in team sports. For comparison papers addressing untrained males who performed maximal exercise on a cycling ergometer were addressed, as that exercise modality requires little technique and is commonly used in physiologic investigations. To broaden the search, studies measuring VO_{2max} and maximal heart rate (HR) were included for calculation of O_2 pulse ($VO_{2\,max}$/HR; Bhambhani et al. 1994) as a surrogate for SV ($n = 735$). Total peripheral resistance (TPR) was calculated by mean arterial pressure divided by allometrically scaled CO if not provided in the study. Reports of athletes' heart dimensions in combination with sport-specific maximal performance are sparse. Thus, to provide an indication of the exercise stimuli associated with cardiac adaptation for each respective sport cardiac variables, recorded at rest, and sport-specific maximal cardiovascular variables are not gathered from the same populations. Variables were allometrically scaled to body weight, rather than to fat-free mass, which is not commonly reported, or to height in order to appreciate variation among athletes (about 60% variation in weight in the included studies vs. 15% variation in height). Also, the difference between body weight and fat-free mass is expected to be small (about 10%) as the included populations are predominantly athletic (Malina 2007; Healy et al. 2014). To increase the validity of the treated data, the mean of variables presented for each population is weighted to the number of participants in each study. The allometric scaling takes into account that LVIDd and WT both represent a distance and thus scaled to $kg^{-0.33}$ (George et al. 1999). On the other hand, metabolism is argued to scale to $kg^{-0.75}$ (West et al. 1997), which apply to human VO_{2max} data (Jensen et al. 2001) and that scaling was also used for CO (Astrand et al. 2003), equally representing a rate (i.e., volume/time), while SV and, indirectly, O_2 pulse as BV that represent a volume were scaled to body weight. In contrast, blood pressure is independent of body size (but related to the vertical distance of the head relative to the heart) (Sandal et al. 2020) and therefore not scaled.

Inclusion criteria were male athletes ($VO_{2max} > 60$ ml kg^{-1} min^{-1} in endurance sports) between 18 and 40 yrs. The control group included age matched males not engaged in regular physical activity ($VO_{2max} \leq 45$ ml kg^{-1} min^{-1}). We searched PubMed (until April 2022) and included references from the lists of previous publications. Four search strings were implemented separately:

1) 'stroke volume', 'cardiac stroke volume', 'stroke volume estimation'
2) 'blood pressure', 'arterial pressure', 'hypertension'
3) 'echocardiography', 'myocardium', 'athlete's heart', 'athletic heart', 'magnetic resonance imaging'
4) 'blood volume', 'plasma volume', 'hemoglobin mass'

and each string included in combination with 'maximal', 'maximum', 'athletes', 'competitive', 'well-trained', 'sport', 'resistance exercise', 'aerobic exercise', 'endurance exercise'.

Statistical analysis

Descriptive data are presented as weighted mean ± standard deviation (SD) of the included means. Differences between variables were compared by one-way Welch ANOVA and post hoc evaluated by the Games-Howell test. A Pearson's correlation identified relationships between physiologic variables and WT and LVIDd. Significantly correlated variables were imputed into a stepwise multiple regression analysis. Statistical significance was set at $P < 0.05$ using SPSS statistics 28 (IBM, Armonk, New York, USA). When sufficient data were not available for statistical analysis, values are narratively described.

Results

The search yielded 2103 studies among which 150 met the inclusion criteria (see online supplementary appendix 1 and 2). There was no difference in age among the studies included for heart dimensions and sport-specific exercise. The populations were of similar age except for the swimmers who were younger ($P = 0.001$). Rowers were taller and heavier than the other athletes and the controls ($P < 0.001$), while runners and cyclists were the lightest ($P < 0.001$).

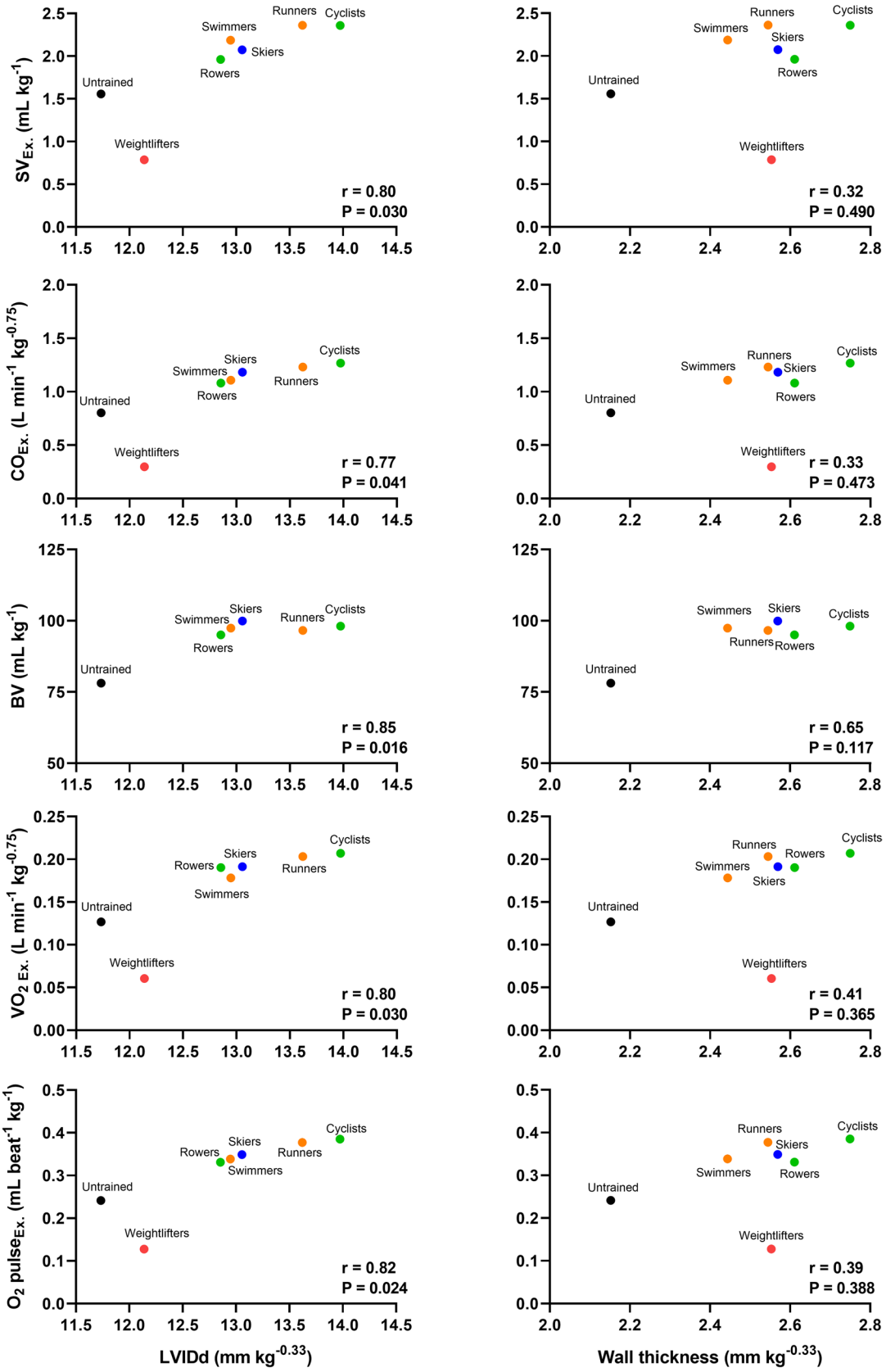
When scaled to body weight, left atrium of endurance athletes was larger (by 6%-17%), with rowers exhibiting the smallest (8.7 ± 0.5 mm $kg^{-0.33}$) and cyclists the largest (9.7 ± 0.9 mm $kg^{-0.33}$) values, compared to controls (8.2 ± 0.8 mm $kg^{-0.33}$), and weightlifters showing a similar dimension to that of the controls (8.2 mm $kg^{-0.33}$). The right atrium was 27% larger in skiers (11.9 ± 0.5 mm $kg^{-0.33}$) and 30% larger in rowers and runners (12.2 ± 0.4 mm $kg^{-0.33}$) compared to controls (9.1 mm $kg^{-0.33}$). Statistical analysis for the atria was not possible due to lack of available values. Similarly, comparison of the right atrium between athletes was limited to skiers, rowers, and runners due to limited number of studies. The right ventricle of endurance athletes (no data available for cyclists) was larger (by 16%-44%, $P = 0.04$), with skiers having the smallest (7.7 ± 1.3 mm $kg^{-0.33}$) and runners the largest values (10.2 ± 1.9 mm $kg^{-0.33}$), while weightlifters demonstrated no elevation (-3%, 6.3 ± 2.2 mm $kg^{-0.33}$) compared to controls (6.5 ± 1.4 mm $kg^{-0.33}$). The BV of all athletes was similarly (within 5%) larger by 20–25% ($P = 0.001$) than in controls with no data available for weightlifters (Table 1).

The diameter of the left atrium was correlated to SV ($r = 0.76$, $P = 0.045$), O_2 pulse ($r = 0.76$, $P = 0.047$), and BV ($r = 0.89$, $P = 0.007$) but only marginally to CO ($r = 0.74$, $P = 0.055$) and VO_{2max} ($r = 0.75$, $P = 0.054$). Also, the diameter of the right ventricle correlated to SV ($r = 0.87$, $P = 0.022$), CO ($r = 0.88$, $P = 0.020$), VO_{2max} ($r = 0.91$, $P = 0.012$), O_2 pulse ($r = 0.90$, $P = 0.015$), and

Table 1 Anthropometric, cardiac, and blood volume data for skiers, rowers, swimmers, cyclists, runners, weightlifters, and untrained

	Skiers	Rowers	Swimmers	Cyclists	Runners	Weightlifters	Untrained
N	117	260	77	177	165	131	173
Age (y)	23 ± 2 *(1-6)*	24 ± 3 *(6-12)*	26 ± 5 *(6,13-16)*	25 ± 7 *(6,17-24)*	27 ± 7 *(6,21,25-29)*	28 ± 4 *(6,26,27,30-36)*	25 ± 3 *(1,13,14,17,24-26,30,35)*
Height (cm)	181 ± 5 *(1,2,4)*	189 ± 3 *(7,10-12)*	179 ± 6 *(13-15)*	180 ± 4 *(17,18,20-24)*	177 ± 4 *(21,26,27)*	177 ± 2 *(26,27,30,33-36)*	178 ± 3 *(1,13,14,17,24,26,30,35)*
Weight (kg)	73 ± 6 *1,2,4,5)*	84 ± 6 *(7,10-12)*	75 ± 4 *(13-15)*	72 ± 4 *(17,18,20-24)*	67 ± 5 *(21,25-27,29)*	90 ± 7 *(26,27,30,33-36)*	75 ± 5 *(1,13,14,17,24-26,30,35)*
LVIDd (mm)	54.5 ± 0.6 *(1-6)*	56.5 ± 2.1 *(6,8-12)*	54.1 ± 1.7 *(6,13-16)*	56.5 ± 2.2 *(6,17,18,20-24)*	54.5 ± 2.7 *(6,21,24-29)*	52.1 ± 2.6 *(6,26,27,30-36)*	49.7 ± 1.9 *(1,13,14,17,24-26,30,35)*
WT (mm)	10.9 ± 0.7 *(1-6)*	11.6 ± 1.1 *(6-12)*	10.2 ± 0.6 *(6,13-16)*	11.1 ± 1.0 *(6,17,18,21-24)*	10.3 ± 0.6 *(6,21,25,26,28,29)*	11.0 ± 1.3 *(6,26,27,30-36)*	9.3 ± 0.7 *(1,13,14,17,24-26,30,35)*
N	154	177	61	246	216	73	319
LA (mm)	39.7 ± 1.2 *(4)*	37.6 ± 2.4 *(12,37-39)*	38.2 ± 3.5 *(40)*	39.9 ± 4.4 *(20-23)*	35.6 ± 0.8 *(21)*	36.0 ± 2.5 *(36)*	34.1 ± 2.4 *(21-23,36-38)*
RVd (mm)	31.6 ± 9.2 *(4,37,41)*	38.3 ± 5.1 *(38,39,42,43)*	34.0 ± 6.0 *(40)*	n/a	41.0 ± 9.1 *(21,41,44,45)*	27.8 ± 10.0 *(36,45,46)*	27.0 ± 6.1 *(21,37,38,41,44-47)*
RA (mm)	48.9 ± 2.9 *(4,41)*	52.7 ± 2.8 *(39,42)*	n/a	n/a	49.1 ± 5.2 *(41)*	n/a	37.6 ± 3.3 *(41)*
BV (ml)	7496 ± 502 *(48-54)*	8265 *(55)*	7471 ± 606 *(48)*	6953 ± 830 *(48,50,56-64)*	6556 ± 884 *(48,53,54,65-67)*	n/a	5685 ± 880 *(52,60,68,69)*

N number of subjects, *LVIDd* diastolic left ventricular internal diameter, *WT* left ventricular posterior WT in diastole, *LA* left atrium, *RVd* right ventricular dimension, *RA* right atrium, *BV* blood volume. Data weighted according to the number of participants in each study. Values are mean ± SD. See online supplementary appendix 1 for references (*numbers*)


SV$_{Ex.}$ (mL kg^{-1})
CO$_{Ex.}$ (L min^{-1} kg$^{-0.75}$)
BV (mL kg^{-1})
VO$_2$ Ex. (L min^{-1} kg$^{-0.75}$)
O$_2$ pulse$_{Ex.}$ (mL beat^{-1} kg^{-1})
LVIDd (mm kg$^{-0.33}$)
Wall thickness (mm kg$^{-0.33}$)
Untrained
Weightlifters
Swimmers
Rowers
Skiers
Runners
Cyclists
r = 0.80
P = 0.030
r = 0.32
P = 0.490
r = 0.77
P = 0.041
r = 0.33
P = 0.473
r = 0.85
P = 0.016
r = 0.65
P = 0.117
r = 0.80
P = 0.030
r = 0.41
P = 0.365
r = 0.82
P = 0.024
r = 0.39
P = 0.388

◂**Fig. 1** Relationship between cardiovascular variables during sport-specific exercise and left ventricular internal diameter (LVIDd) and WT (WT) at rest scaled to body weight for skiers, rowers, swimmers, cyclists, weightlifters, and untrained. $SV_{Ex.}$ stroke volume during exercise, $CO_{Ex.}$ cardiac output during exercise, *BV* blood volume at rest, $VO_{2\ Ex}$ oxygen uptake during exercise. Colors indicate differences in peak dynamic and static components achieved during competition based on the classification of Levine et al. (2015). red; low dynamic and high static, blue; high dynamic and low to moderate static, orange; high dynamic and moderate static, green; high dynamic and high static, black; untrained

BV ($r = 0.89$, $P = 0.017$), while there was a lack of data for the right atrium.

All athletes, except the weightlifters had larger LVIDd than the untrained ($P < 0.001$) with cyclists exhibiting the largest values (15% compared to controls; Fig. 1). The WT was up to 24% larger in athletes than in the untrained ($P < 0.001$) with no significant difference among groups. During exercise, weightlifting required the lowest VO_2 ($P < 0.001$), while cyclists exhibiting the highest VO_{2max} (48% larger than the untrained group). Conversely, weightlifting developed a ~ 70% higher mean arterial pressure ($P < 0.001$) than all others (Table 2 and Fig. 2). During exercise, weightlifting resulted in the lowest SV ($P < 0.001$), while the other athletes demonstrated 23%-41% larger SV than the untrained. Similarly, O_2 pulse and CO of endurance athletes was up to ~ 45% ($P < 0.001$) larger than for the untrained. Thus, when scaled to body weight, the cyclists and runners demonstrated larger volume load and LVIDd, while cyclists exhibited the largest WT (Fig. 1). There was a correlation between LVIDd and WT ($r = 0.79$, $p = 0.04$). The LVIDd was correlated to SV ($r = 0.80$), CO ($r = 0.77$), $VO_{2\ max}$ ($r = 0.80$), O_2 pulse ($r = 0.82$), and BV ($r = 0.85$, all $P < 0.05$), but weightlifters were not included in the evaluation of BV due to lack of data. In contrast, there was no correlation between LVIDd and HR ($r = 0.69$), systolic ($r = -0.44$), diastolic ($r = -0.62$), mean arterial pressure ($r = -0.60$), or TPR ($r = -0.61$, $P > 0.05$). Also, there was no correlation between WT and any variable. A stepwise multiple regression revealed BV as the independent predictor for LVIDd, $F(1,5) = 18.663$, $P = 0.008$, while no variable predicted WT, $F(4,2) = 1.1321$, $P = 0.47$.

Discussion

Here we describe heart dimensions at rest in male athletes while taking body weight into account to eliminate the bias introduced by the wide variability in the size of athletes (Khosla 1978, 1983). Allometric scaling of published variables for the investigated athletic groups demonstrated increased WT and LVIDd (except for weightlifters) providing agreement with the considerations by Morganroth et al. (1975). Furthermore, data are provided for sport-specific exercise to supplement the classifications of sports (Mitchell et al. 2005; Levine et al. 2015). The LVIDd was related to VO_{2max}, SV, O_2 pulse, CO, and BV ($r = \sim 0.9$, $P < 0.05$), while there was no relationship between these variables and WT ($P > 0.05$). Together, these observations relate athletic endurance performance to an effect on LVIDd, while an increase in WT may be secondary to the effort the heart needs to generate blood pressure in face of an enlarged LVIDd. In support, we found a correlation between WT and LVIDd among the populations as expected according to Laplace's law although that would suggest only half the expansion of WT than of LVIDd without taking the contribution of ventricular "twisting" into account (Fig. 3).

Concentric vs. eccentric hypertrophy

Both endurance and "resistance" athletes had a larger WT than the control group, with WT not being significantly different among athletes. In contrast, LVIDd was not significantly different between resistance athletes and controls, while LVIDd was larger in endurance athletes than resistance athletes, in accordance with the "concentric" hypertrophy suggested by Morganroth et al. (1975) and others (Macfarlane et al. 1991; Vinereanu et al. 2002). Pluim et al. (2000) found support for concentric hypertrophy in resistance athletes expressed as an increased wall to chamber ratio, i.e., relative WT. Here resistance athletes had a larger relative WT (0.42) compared to controls (0.37) and endurance athletes (ranging from 0.37–0.41) but all values remained within the normal range (Lang et al. 2005). The cardiac adaptation of resistance-trained athletes is, however, debated (Haykowsky et al. 2000, 2018; Wernstedt et al. 2002). Haykowsky et al. (2001) found transmural pressure to equalize during resistance exercise, meaning that wall stress, or afterload, is not greater than at rest, making it unclear why the heart of resistance athletes develops concentric hypertrophy. Other potential training and stimulus-specific functional remodeling of the resistance athletes' LV may be involved as they seem to maintain SV during resistance exercise (Dawkins et al. 2020). The effect of resistance training on WT could also be due to the use of anabolic steroids (Urhausen et al. 2004), but we included only studies addressing stated non-users.

The whole heart

Primarily LV remodeling is investigated, while data regarding the right ventricle and the atria are limited, especially in resistance athletes (Saunders et al. 2022). We found the right ventricle and the atria are larger in endurance athletes compared to untrained, while they were not enlarged in resistance athletes as previously reported (D'Andrea et al. 2013;

Table 2 Anthropometric and cardiovascular variables for skiers, rowers, swimmers, cyclists, runners, weightlifters, and untrained

	Skiers	Rowers	Swimmers	Cyclists	Runners	Weightlifters	Untrained
N	110	157	100	147	178	119	221
Age (y)	24 ± 2 *(1–11)*	23 ± 3 *(12–26)*	19 ± 2 *(27–36)*	24 ± 4 *(37–47)*	25 ± 4 *(3,48–60)*	25 ± 3 *(61–71)*	25 ± 4 *(33,35,37,44,51,52,58,72–78)*
Height (cm)	181 ± 4 *(1–11)*	188 ± 4 *(12–21,23–26)*	181 ± 3 *(27–36)*	179 ± 3 *(40–47)*	178 ± 3 *(3,48,49,52–60)*	179 ± 3 *(63,64,66–69,71)*	179 ± 3 *(33,35,44,51,52,58,74–78)*
Weight (kg)	77 ± 4 *(1–11)*	84 ± 7 *(12–26,79)*	74 ± 4 *(27–36)*	70 ± 3 *(37–47)*	65 ± 4 *(3,48,49,52–60)*	82 ± 8 *(63,64,66–69,71)*	77 ± 4 *(33,35,37,44,51,52,58,74–78)*
HR (bpm)	188 ± 10 *(1–11)*	187 ± 5 *(12,13,17–19,21–26,79)*	183 ± 8 *(27–34,36)*	186 ± 5 *(37–47,80)*	186 ± 5 *(3,48–60)*	151 ± 17 *(61–71)*	184 ± 9 *(33,37,44,51,52,58,72–78)*
SYS (mmHg)	120 ± 15 *(1)*	194 ± 12 *(12–14)*	n/a	195 ± 1 *(38,39)*	181 ± 1 *(50,51)*	320 ± 33 *(62,63,65,70,81)*	186 ± 20 *(51,73,77,78)*
DIA (mmHg)	58 ± 4 *(1)*	57 ± 25 *(13,14)*	n/a	67 ± 2 *(38,39)*	68 ± 4 *(50)*	240 ± 33 *(62,63,65,70,81)*	82 ± 5 *(73,77,78)*
MAP (mmHg)	100 ± 13 *(1–3)*	114 ± 7 *(12–14)*	132 ± 12 *(29)*	114 ± 8 *(38,39,46)*	110 ± 11 *(3,50)*	239 ± 41 *(62,63,65,70,81)*	116 ± 11 *(72,73,77,78)*
SV (ml min^{-1})	159 ± 19 *(1,3)*	165 ± 10 *(13)*	162 ± 18 *(29,33)*	166 ± 23 *(37–39,80)*	153 ± 24 *(3,48,49,52,58)*	64 ± 15 *(64,68,69)*	120 ± 7 *(33,37,44,52,58,72,73,76)*
CO (l min^{-1})	31 ± 3 *(1,3)*	30 ± 2 *(13)*	28 ± 3 *(29,33,35)*	31 ± 4 *(37–40,44,46,80)*	28 ± 4 *(3,48,49,52,58)*	8 ± 2 *(65,68,69)*	21 ± 3 *(33,35,37,44,52,58,72,73,76)*
VO_2 (l min^{-1})	5.0 ± 1.0 *(1–11)*	5.3 ± 0.7 *(12,13,15–26,79)*	4.5 ± 07 *(28–32,34–36)*	5.0 ± 0.3 *(38–47,80)*	4.7 ± 0.5 *(3,48,49,52–60)*	1.6 ± 0.3 *(66,67)*	3.3 ± 0.4 *(33,35,37,44,52,58,72–76,78)*
O_2 pulse (ml $beat^{-1}$)	27 ± 5	28 ± 4	25 ± 3	27 ± 2	25 ± 3	10 ± 1	19 ± 2

N number of subjects, *HR* heart rate, *SYS* systolic pressure, *DIA* diastolic pressure, *MAP* mean arterial pressure, *SV* stroke volume; CO, cardiac output; VO_2, pulmonary oxygen uptake. Data weighted according to the number of participants in each study. Values are mean ± SD. See online supplementary appendix 2 for references (*numbers*)

Utomi et al. 2015). D'Ascenzi et al. (2017) found about one-third of Olympic athletes present increased right ventricular dimensions that even exceed the reference values. The adaptations in endurance athletes seem balanced as there is a correlation between LVIDd, the left atrium (Nistri et al. 2011), and right ventricle enlargement (Steding et al. 2010) and furthermore the ratio of left atrial volume and LVIDd does not differ among endurance and resistance athletes and controls (Kooreman et al. 2019). The effect of endurance training on the enlargement of cardiac cavities could be mediated through the volume overload as both ventricles and left atrium correlate to VO_{2max} (Osborne et al. 1992; Barbier et al. 2006; Steding et al. 2010) and our findings add correlations to SV, O_2 pulse, CO, and BV.

Allometric scaling

In 23 of the reviewed studies (Appendix 1) no scaling was applied for LVIDd or WT and it therefore remains uncertain to what extent it is an influence of body mass or training that is reported. Alternatively, when WT and LVIDd are related to a measure of body surface area, the effect of training on the heart is underestimated for large athletes. Among these studies, Spirito et al. (1994) also took age and sex into consideration. It was argued that an influence of training on the heart takes time and may be therefore why the young swimmers demonstrate only small deviations. The difficulty encompassed in scaling variables also applies to fat-free mass (Urhausen et al. 1989, 2004), which further reduces the expression of heart size in the tall Olympic athlete (Khosla 1983). George et al. (1999) scaled the description of the heart to a "length" that is relevant for both WT and LVIDd. Yet, the paper presents several other scaling measures of the heart. Another paper relates LVIDd to the square root of body surface area (Venckunas et al. 2011) and thereby to a "length". This report, however, divided LV posterior wall plus intra-ventricular septum thickness by LVIDd for evaluation of WT and thus, by dividing a "length" by a "length" eliminated scaling in that regard.

Also, the exponents used to scale other variables than those applied for the heart vary. The most often scaled variable may be CO expressed as cardiac index, i.e., CO divided by an estimate of body surface area (often length squared). Alternatively, CO might be related to $kg^{-0.67}$ (von Döbeln 1957) while we applied $kg^{-0.75}$ according to the considerations by West et al. (1997). Similarly, $VO_{2\ max}$ was scaled to $kg^{-0.75}$ (Jensen et al. 2001). Yet, accepting the differences in body shape among individuals (Olds 2008), we assume that application of body weight for scaling accounts for much of this variation.

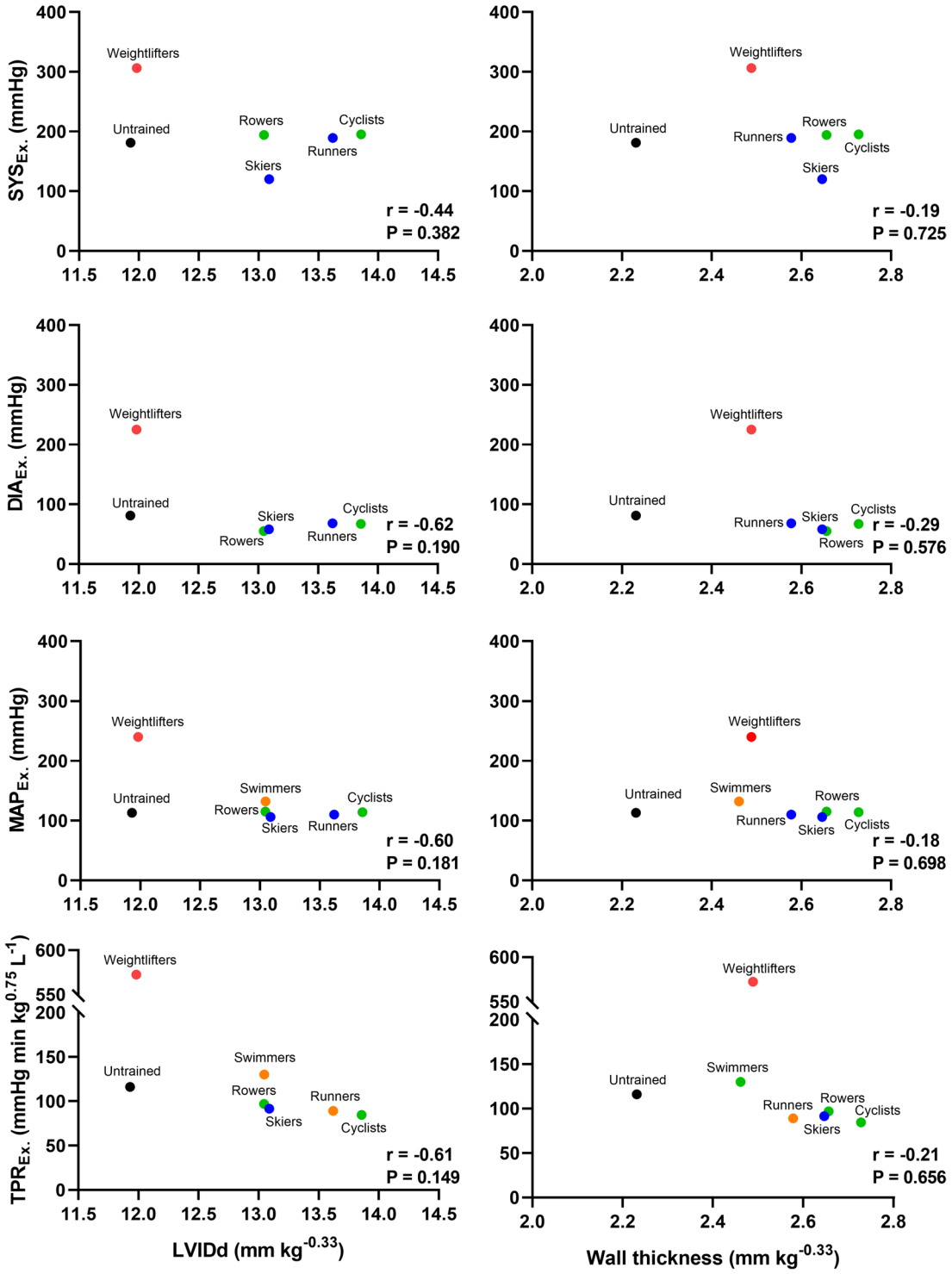


Fig. 2 Relationship between cardiovascular variables during sport-specific exercise and left ventricular internal diameter (LVIDd) and wall thickness (WT) at rest scaled to body weight for skiers, rowers, swimmers, cyclists, weightlifters, and untrained. $SYS_{Ex.}$ systolic pressure during exercise, $DIA_{Ex.}$ diastolic pressure during exercise, $MAP_{Ex.}$ mean arterial pressure during exercise, TPR_{Ex} total peripheral resistance during exercise. Colors indicate differences in peak dynamic and static components achieved during competition based on the classification of Levine et al. (2015). red; low dynamic and high static, blue; high dynamic and low to moderate static, orange; high dynamic and moderate static, green; high dynamic and high static, black; untrained

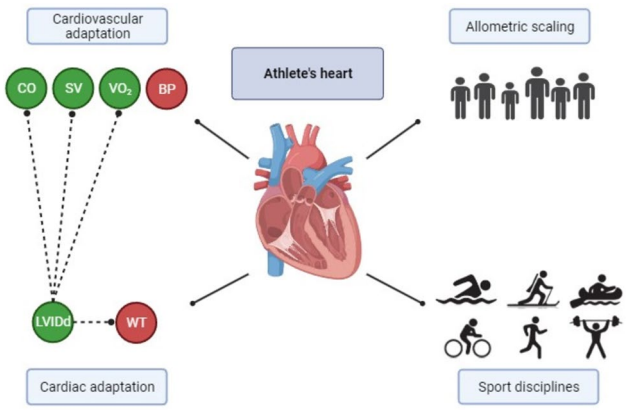


Fig. 3 The cardiovascular load depends on the types of athletic endeavor. Investigating sport disciplines including different levels of intensity of dynamic and static exercise provides a framework to evaluate how the maximal cardiovascular load of each sport discipline provokes cardiac remodeling over time. To account for the confounding effects of body size variables for athletes in different sport disciplines are exposed to allometric scaling. The proposed adaptation of expanded blood volume, with concomitant increase in stroke volume (SV) and cardiac output (CO), enlarges the left ventricular internal diastolic diameter (LVIDd), while an increased left ventricular wall thickness (WT) may primarily reflect the effort the heart has to generate blood pressure in face of the enlarged SV and LVIDd. Created with BioRender.com. CS and NHS conceived and designed the research. CS collected and analyzed the data. CS and NHS interpreted the results. CS and NHS drafted the manuscript with all the authors involved in the revision and approval of the final version of the manuscript

Limitations

The data for the heart and exercise were not from the same population as studies evaluating the athlete's heart often is cross-sectional and does not include sport-specific cardiovascular measurement. Also, it is standard to evaluate cardiac dimensions at rest and exercise measurements during all the included endurance sports are limited by e.g., posture and movement artifacts. Furthermore, we did not discriminate between athletes participating in different disciplines within the same sport, e.g., sprinters and long-distance runners, or power lifters, bodybuilders, and weightlifters. Likely, there are differences both between cardiac morphology and cardiovascular response to exercise in different resistance-trained athletes (Falkel et al. 1992; Pelliccia et al. 1993). A meta-analysis approach could have been adopted, to limit, e.g., publication bias, yet it would have been difficult to implement due to the divergent purposes of the included studies. Thus, this study might be influenced by publication bias but such influence is deemed as negligible, as the purpose of the study was to exemplify how the cardiovascular load of different sports over time could provoke different cardiac adaptations. Also, we did not include female athletes due to lack of sport-specific cardiovascular data. Furthermore, the primary activity of each sport/event is likely supplemented with "specific" training of muscle groups including, e.g., weight lifting and running/cycling, resulting in a mixed stimulus. Similarly, we could not evaluate whether cardiac measurements were established after a period of increased training volume. The CO and VO_2 during weightlifting were recorded during submaximal exercise (~ 70% one repetition maximum), while BP of skiers during 76% VO_{2max} and therefore are not representative of maximal work. We used an allometric approach relating cardiac and hemodynamic variables to body weight but scaling to other anthropometric variables could be applied. Finally, scaling was based on mean values and we acknowledge the need for individual values to account for heterogeneity among subjects.

Conclusion

Here we address the importance of allometric scaling when comparing the heart of male athletes of different body size. We provide an indication of what exercise stimuli results in remodeling in athletes of different sports and correlate LV dimensions to BV. With allometric scaling to body weight, the lighter athletes possess a higher CO and LVIDd, while LVIDd is similar among controls and weightlifters. The enlarged LVIDd of athletes is correlated to BV, while an increase in WT may reflect the effort of the heart to generate blood pressure in face of an enlarged SV and LVIDd.

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Author contributions CS and NHS conceived and designed the research. CS collected and analyzed the data. CS and NHS interpreted the results. CS and NHS drafted the manuscript with all the authors involved in the revision and approval of the final version of the manuscript.

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Data availability All data discussed in this review comes from published work.

Declarations

Conflict of interest The authors declare that they have no competing interests.

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